

Akhilesh Venkataramanan

Machine Learning Engineer

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SUMMARY:

Machine Learning Engineer with over 4 years of focused experience in deep learning, computer vision, and natural language processing. Demonstrates proficiency in Python, SQL, and C++ and deep expertise in TensorFlow, Keras, PyTorch, and Scikit-learn. Develops and deploys complex neural networks, including CNN, RNN, LSTM, GAN, and Transformers. Utilizes advanced NLP techniques with NLTK, SpaCy, BERT, GPT-3, and Hugging Face Transformers. Manages model training and deployment on AWS, Azure, and GCP to ensure scalability. Excels in feature engineering, model tuning, validation processes, and statistical analysis. Applies a solid mathematical foundation to enhance data visualization with Tableau, Power BI, and Plotly.

SKILLS:

- Methodology:** SDLC, Agile, Waterfall
- Programming Language:** Python, SQL, C++
- IDE's:** PyCharm, Jupyter Notebook, Visual Studio Code
- Machine Learning Frameworks and Libraries:** TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, LightGBM
- Data Processing and Analysis:** NumPy, Pandas, SciPy, Matplotlib, Seaborn
- Deep Learning:** Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory Networks (LSTM), Generative Adversarial Networks (GAN)
- Natural Language Processing (NLP):** NLTK, SpaCy, BERT, GPT-3, Transformer models
- Cloud Platforms:** Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP)
- Development Tools:** Jupyter Notebooks, Git, Docker, Kubernetes, Apache Airflow
- Model Deployment:** Flask, Fast API, TensorFlow Serving, MLflow
- Data Visualization:** Tableau, Power BI, Plotly, D3.js
- Mathematics and Statistics:** Linear Algebra, Calculus, Probability, Statistics
- Other Skills:** Feature Engineering, Model Selection and Evaluation, Hyperparameter Tuning, Cross-Validation A/B Testing, Data Cleaning and Preprocessing, Critical Thinking, Problem-Solving, Communication Skills, Team Collaboration
- Operating Systems:** Windows, Linux

EDUCATION:

- Master of Robotics and Autonomous Systems, Artificial Intelligence** Aug 2022 – May 2024
Arizona State University, Tempe, AZ
- Coursework:** Artificial Intelligence, Artificial Neural Computation, Perception in Robotics, Statistical ML, Embedded ML, Data Processing at Scale
- Bachelor of Engineering in Electronics and Computer Engineering** Aug 2015 – June 2019
RNS Institute of Technology, Bengaluru, India

EXPERIENCE:

INTUIT, AZ | July 2023 – Current | Machine Learning Engineer

- Spearheaded development and optimization of deep learning models leveraging TensorFlow, Keras, and PyTorch, achieving a 25% increase in predictive accuracy across multiple use cases.
- Built and fine-tuned advanced neural networks (CNN, RNN, LSTM, GAN), improving image recognition and NLP tasks by 20% through robust data augmentation and architecture optimization.
- Deployed scalable computer vision applications using OpenCV and Dlib, improving video analysis performance by 30% and reducing latency.

- Implemented NLP models with BERT and GPT-3, enhancing sentiment analysis accuracy by 15% via ensemble learning and custom model stacking techniques.
- Deployed models on cloud platforms (AWS, Azure, GCP), achieving 99.9% uptime through automated scaling and a 35% reduction in deployment time using Flask, FastAPI, and TensorFlow Serving.
- Designed interactive dashboards using Tableau, Power BI, and Plotly, boosting stakeholder engagement by 30% with real-time visualizations.

Trigent Software Pvt Ltd, Bengaluru| Nov 2021 – June 2022 | Data Scientist - Computer Vision

Tech Stack: Python, OpenCV, TensorFlow/Keras, PyTorch, Git, version control

- Executed object detection models using OpenCV, Dlib, and ImageAI, incorporating deep learning algorithms like Faster R-CNN, YOLO, SSD, and Mask R-CNN to achieve superior accuracy and real-time performance.
- Designed image and video processing pipelines, integrating FFmpeg, Scikit-Image, and moviepy, and utilized Albumentations, AugLy, and imgaug for robust data augmentation, enhancing generalization.
- Constructed custom CNN architectures and applied transfer learning techniques using models like ResNet, VGGNet, and Inception for refined image classification and segmentation; streamlined data labeling and preprocessing using tools such as LabelImg, CVAT, and VGG Image Annotator.
- Optimized real-time computer vision applications and developed advanced 3D vision solutions using TensorRT, NVIDIA DeepStream, PCL, and Open3D, expanding capabilities in processing complex 3D data.
- Developed and refined machine learning models for image recognition, elevating model accuracy by 20%.
- Crafted data pipelines and preprocessing scripts, reducing preparation time by 30%; performed exploratory data analysis and visualization with NumPy, Pandas, Matplotlib, and Seaborn.
- Produced detailed visualizations with Tableau, Power BI, Plotly, and D3.js to communicate findings, effectively boosting data-driven decision-making by 25%.
- Engineered and validated models, conducting feature engineering, model selection, hyperparameter tuning, cross-validation, and A/B testing, improving performance by 20% and enhancing cross-functional collaboration by 15%.

Flux Auto India Pvt Ltd, Bengaluru| July 2019 – Aug 2021 | Deep Learning Engineer

Tech Stack: Python, OpenCV, TensorFlow/Keras, PyTorch, Git, Amazon Web Services (EC2 and S3), Ubuntu/Linux, CUDA, GPU, ROS

- Led the development of deep learning models for keypoint detection and pose estimation, achieving a mean Average Precision (mAP) score of over 0.9.
- Engineered comprehensive pipelines for a research paper, including data processing, network architecture, loss functions, and training on an AWS EC2 instance for 3D object detection and depth estimation.
- Deployed a fully convolutional network for drivable area segmentation, attaining more than 60 frames per second and a mean Intersection over Union (mIOU) score of over 0.92; integrated object detection and semantic segmentation into a multi-task model, enhancing performance.
- Implemented Pix2Pix GAN for effective image-to-image translation, achieving over 80% accuracy; explored and implemented convolutional LSTM networks for lane segmentation on time sequence frames, earning a mean IOU score of over 0.8.
- Built and optimized deep learning models using TensorFlow, Keras, PyTorch, and Scikit-learn, specializing in neural networks such as CNNs, RNNs, LSTMs, GANs, and Transformer models (e.g., BERT, GPT-3).
- Deployed deep learning applications on cloud platforms like AWS, Microsoft Azure, and GCP, utilizing Docker, Kubernetes, and Apache Airflow for scalable training and efficient deployment.
- Managed data processing and analysis using Python libraries to ensure data quality and readiness for training.
- Enhanced model performance through meticulous feature engineering, hyperparameter tuning, and cross-validation; ensured robust deployment using Flask, FastAPI, TensorFlow Serving, and MLflow.

PUBLICATIONS:

- Drone to Obstacle Distance Estimation Using YOLOv3 Network and Mathematical Principles | AICECS 2021 | [Link](#)
- Encoding web-based data for efficient storage in Machine Learning applications | ICInPro2019 | [Link](#)

- Sentiment analysis of restaurant reviews using various Machine Learning techniques | ICERECT2018 | [Link](#)
- Sales forecasting of Retail Stores using Machine Learning Techniques | CSITSS2018 | [Link](#)

PROJECTS:

Human Pose Estimation on Raspberri Pi 4, Embedded Machine Learning Course Project

Nov 2023

- Trained and Evaluated High Resolution Net using PyTorch to recognize 17 keypoints over human body, that exhibited 0.87 mAP score.
- Deployed on Raspberry Pi 4 edge device (embedded system) using TFLite framework.

Dexterous Hand Manipulation Model using Deep Reinforcement Learning, Perception in Robotics Course Project April 2023

- Simulated the object relocation task of the Robotic Arm on the MuJoCo environment by designing a Multi-Layer Perceptron policy.
- Computed Demo Augmented policy gradient & performed Proximal Policy Optimization that accelerated the success rate convergence by 50%.

Sketch-to-Face synthesis – Image to Image translation, Artificial Neural Computation Course Project | [GitHub](#)

Dec 2022

- Trained Style GAN in PyTorch with 18 different styles to generate real human face images by providing hand-sketched images as the input.
- Generated synthetic dataset by utilizing OpenCV to create synthetic hand sketches from CelebAHQ dataset.

Automatic Helmetless Rider Detection using Deep Learning, Undergrad Final year Project | [YouTube](#)

May 2019

- Collected custom dataset of real-world traffic, trained YOLOv3 model for person and motorcycle detection and designed an image classifier for helmet classification.
- Generated synthetic data for license plate detection training and performed contour detection for license plate recognition.

ACHIEVEMENTS

- Winner of the BITES state-level competition 2019 for the project "Automatic Helmetless Rider Detection using Deep Learning" | [Link](#)
- Best Software project winner, FINAL YEAR project expo 2019 for "Automatic Heltmetless Rider Detection using Deep Learning" | [Link](#)
- Secured Best Paper award in the 3rd ICERECT 2018 for "Sentiment Analysis of Restaurant Reviews using various Machine Learning techniques" paper | [Link](#)